

Product Line Profitability & Margin Performance Analysis

A Data-Driven Study of Nassau Candy Distributor's Transaction Records

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Abstract — Sales volume alone does not indicate which products are financially valuable to a distributor. This study analyzes 9,881 cleaned transaction records covering 15 products across three divisions from Nassau Candy Distributor to identify which products and divisions actually drive profit, where margins are weak, and how concentrated financial performance is across the catalog. After validating and cleaning the source data, product- and division-level profitability metrics were computed and a Pareto concentration analysis was performed. The results show a business whose \$138,284 in total sales and \$91,076 in gross profit (an overall 65.9% margin) are dominated by a single division and a small subset of products, with one clear margin-risk outlier identified for further review.

1. Introduction

Nassau Candy Distributor records every order line it processes — product, division, region, units sold, sales revenue, cost, and gross profit — but this raw transaction log does not by itself answer the question that matters most to the business: which products are actually worth selling. A product can move a large volume of units and still be a weak contributor to profit if its margin is thin, while a lower-volume product can be quietly efficient. This paper turns Nassau Candy's raw transaction data into a structured profitability analysis that separates sales volume from financial value.

1.1 Business Problem

Sales volume alone does not tell Nassau Candy which products are financially valuable. Some products may generate high sales but low profit, carry high costs against weak margins, or generate strong profit despite lower sales volume. Without a profitability-first view, sourcing, pricing, and product-rationalization decisions risk being driven by the wrong signal.

1.2 Objectives

- Identify which products generate the most gross profit, and which have the highest gross margins.
- Identify high-sales products with comparatively weak margins.
- Compare divisions on revenue, profit, and margin performance.
- Quantify how concentrated revenue and profit are across the product catalog (Pareto analysis).
- Flag products with cost/margin risk that may warrant pricing, sourcing, or cost-reduction review.

This is a data analytics / business analytics study. Machine learning was deliberately not introduced: the dataset (15 products, 3 divisions) and the stated objective are best served by transparent profitability metrics, threshold-based segmentation, and concentration analysis rather than predictive modeling.

2. Data & Methodology

2.1 Dataset

The source data is a single-sheet Excel workbook of 10,194 order line items across 18 columns, covering order and shipping metadata, customer location, product identifiers (Product ID, Product Name, Division), and financial fields (Sales, Units, Cost, Gross Profit). The catalog spans 15 products across three divisions — Chocolate, Other, and Sugar — sold across four regions (Pacific, Atlantic, Interior, Gulf) in the United States and Canada, with order dates ranging from January 2024 to December 2025.

2.2 Data Cleaning

- **Completeness:** no missing values were found in any of the 18 source columns.
- **Duplicates:** 313 rows were found to be identical across every field except the surrogate Row ID (≈2.5% of total Sales dollars) and were removed, keeping the first occurrence.
- **Order Date:** stored as a mix of native Excel dates and DD-MM-YYYY text strings within the same column; both were confirmed to represent the same format and parsed into one consistent datetime field.

- **Ship Date:** found to be unreliable — parsed values fall in 2026–2030, years after the corresponding Order Date — and was excluded from all analysis rather than used or silently trusted.
- **Gross Profit:** already present in the source data; validated as exactly Sales – Cost for every row (maximum floating-point residual $\approx 7 \times 10^{-15}$) and therefore not recreated.
- **Numeric validity:** no negative, zero, or otherwise invalid Sales, Units, Cost, or Gross Profit values were present.
- **Scope check:** the original project brief referenced factory coordinates and shipping-route analysis; this dataset contains no factory, coordinate, or route data, so that thread was left out rather than fabricated.

Cleaning produced 9,881 clean transaction rows used for all analysis in this paper.

2.3 Metric Definitions

Metric	Formula
Gross Profit	Sales – Cost (validated, not recreated)
Gross Margin %	Gross Profit / Sales × 100
Profit Per Unit	Gross Profit / Units
Revenue Contribution %	Product Sales / Total Sales × 100
Profit Contribution %	Product Gross Profit / Total Gross Profit × 100
Margin Volatility	Std. deviation of a product's monthly gross margin

Product segmentation (High Performing / Margin Risk / Potential Opportunity / Requires Review) is based on each product's Sales and Gross Margin relative to the **median** across the catalog — a transparent, data-driven threshold rather than an arbitrary cutoff. These segments are treated as review flags for further investigation, not automatic business decisions.

3. Results

3.1 Company-Level Overview

KPI	Value
Total Sales	\$138,284.12
Total Gross Profit	\$91,076.41
Overall Gross Margin	65.86%
Total Units Sold	37,667
Products	15

3.2 Product-Level Profitability

Table 2 ranks all 15 products by total Sales, alongside Gross Profit and Gross Margin %. The five Wonka Bar chocolate SKUs occupy the top five positions on both Sales and Gross Profit.

Product	Division	Sales (\$)	Gross Profit (\$)	Margin (%)
Wonka Bar - Triple Dazzle Caramel	Chocolate	27,648.75	18,063.85	65.3
Wonka Bar -Scrumdiddlyumptious	Chocolate	27,028.80	18,770.00	69.4
Wonka Bar - Milk Chocolate	Chocolate	26,256.75	17,046.69	64.9
Wonka Bar - Fudge Mallows	Chocolate	24,361.20	16,240.80	66.7
Wonka Bar - Nutty Crunch Surprise	Chocolate	22,897.89	16,336.89	71.3
Lickable Wallpaper	Other	7,860.00	3,930.00	50.0
Kazookles	Other	1,205.75	92.75	7.7
Wonka Gum	Other	597.50	310.70	52.0
Everlasting Gobstopper	Sugar	130.00	104.00	80.0
Fizzy Lifting Drinks	Sugar	78.75	47.25	60.0
Hair Toffee	Sugar	76.50	59.50	77.8
SweeTARTS	Sugar	61.50	28.70	46.7
Laffy Taffy	Sugar	53.73	33.48	62.3
Nerds	Sugar	15.00	7.00	46.7
Fun Dip	Sugar	12.00	4.80	40.0

Table 2. Product profitability, ranked by Sales.

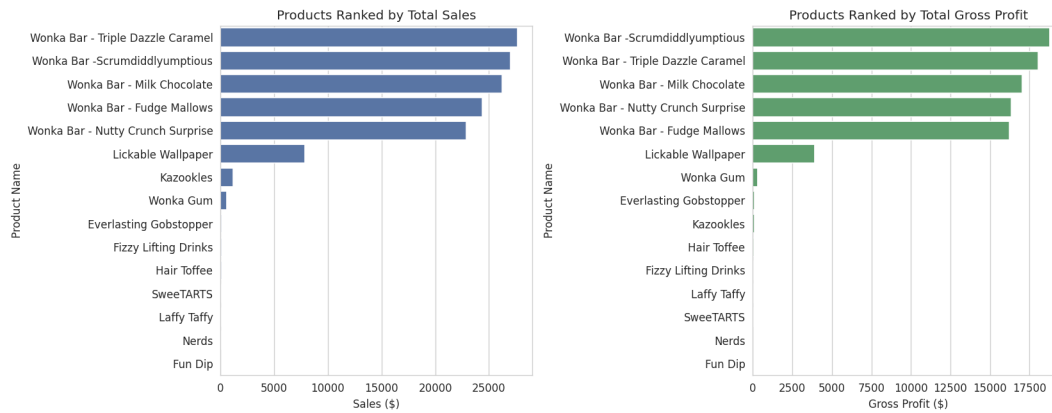


Figure 1. Products ranked by total Sales (left) and total Gross Profit (right).

Outside the five Wonka Bars, **Kazookles** (Other division) stands out: it has meaningful sales (\$1,205.75) but only a 7.7% gross margin — the lowest in the entire catalog by a wide margin, and well below the next-lowest product. **Everlasting Gobstopper** (Sugar) has the single highest gross margin at 80.0%, but with only \$130 in total sales, it is a marginal contributor to total company profit despite its efficiency.

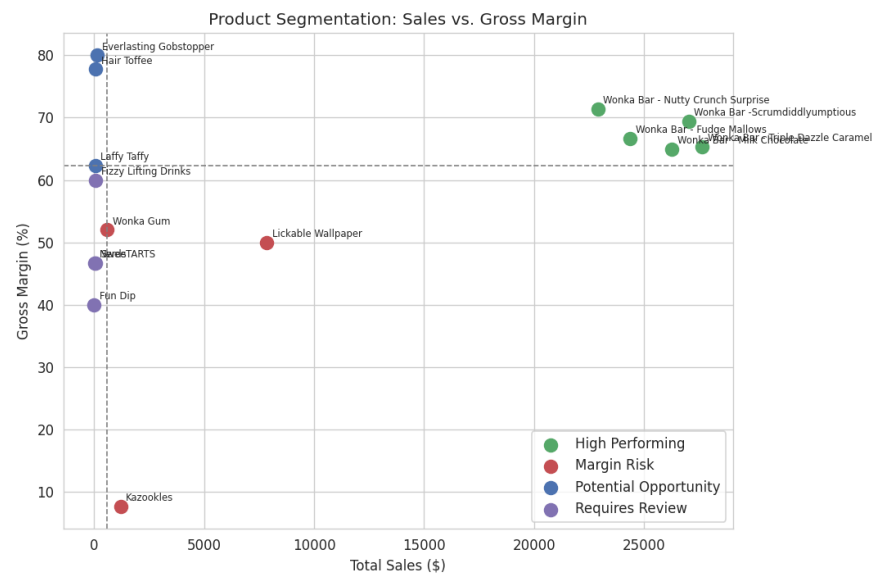


Figure 2. Product segmentation by Sales vs. Gross Margin, split at the catalog median on each axis.

3.3 Division Performance

Division	Sales (\$)	Gross Profit (\$)	Avg Margin (%)	Revenue Contrib. (%)	Profit Contrib. (%)
Chocolate	128,193.39	86,458.23	67.4	92.7	94.9
Other	9,663.25	4,333.45	44.8	7.0	4.8
Sugar	427.48	284.73	66.6	0.3	0.3

Table 3. Division-level performance.

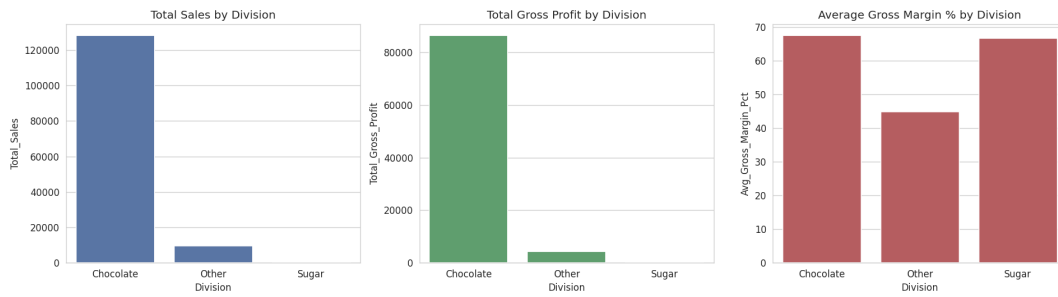


Figure 3. Sales, Gross Profit, and Gross Margin % by division.

Chocolate accounts for 92.7% of revenue and 94.9% of gross profit — its profit contribution exceeds its revenue contribution, meaning it converts sales to profit slightly more efficiently than the catalog average. Other's profit contribution (4.8%) trails its revenue contribution (7.0%), consistent with Kazookles pulling down that division's overall margin. Sugar contributes a negligible 0.3% of both revenue and profit, despite individual Sugar products often having strong per-unit margins.

3.4 Profit Concentration (Pareto Analysis)

A Pareto analysis was performed separately on revenue and gross profit, sorting products in descending order and tracking cumulative contribution. **5 of 15 products (33.3%) generate the first 80% of total revenue**, and **5 of 15 products (33.3%) generate the first 80% of total gross profit**. In both cases, those products are the five Wonka Bar chocolate SKUs.

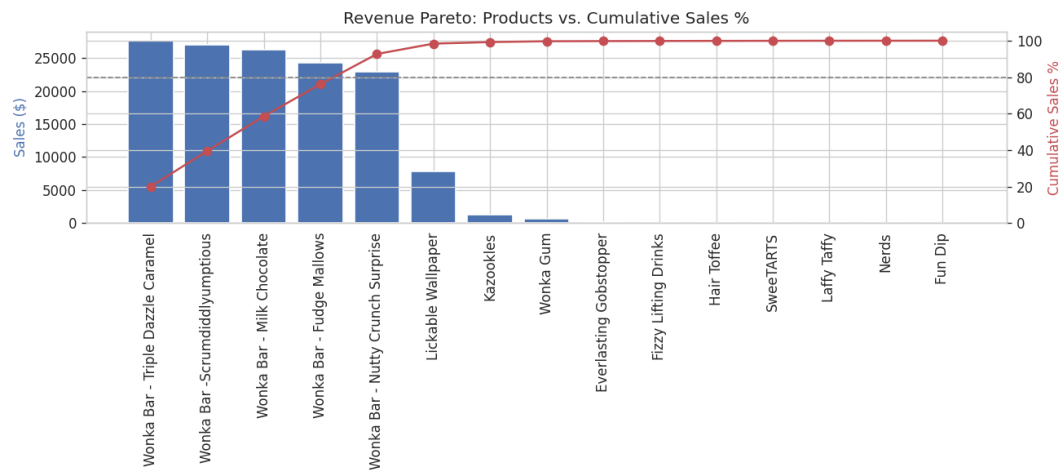


Figure 4. Revenue Pareto — cumulative Sales % by product, sorted descending.

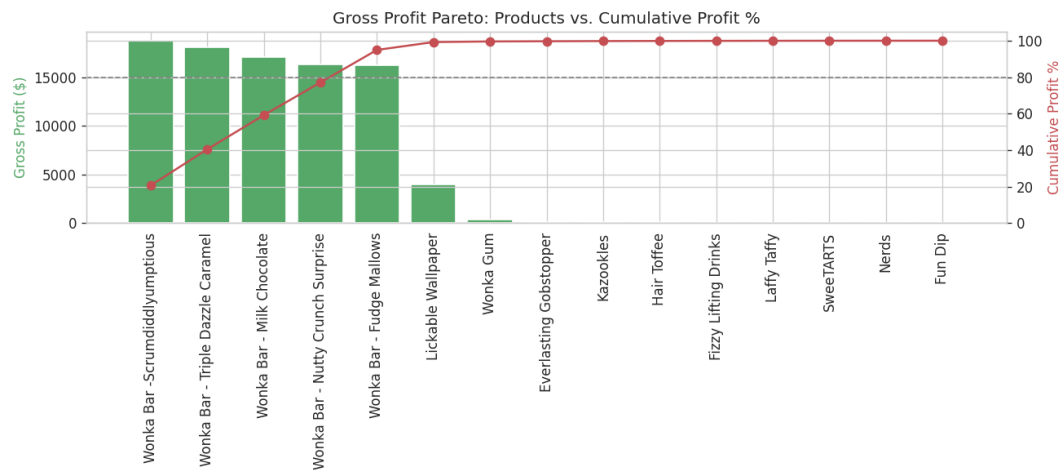


Figure 5. Gross Profit Pareto — cumulative Gross Profit % by product, sorted descending.

This concentration indicates a dependency risk: a third of the catalog drives the large majority of the company's financial performance, while most of the remaining products — especially in the Sugar division — contribute only marginal dollars. A disruption to demand, cost, or supply for any of the five core products would have an outsized effect on total company revenue and profit.

3.5 Cost & Margin Diagnostics

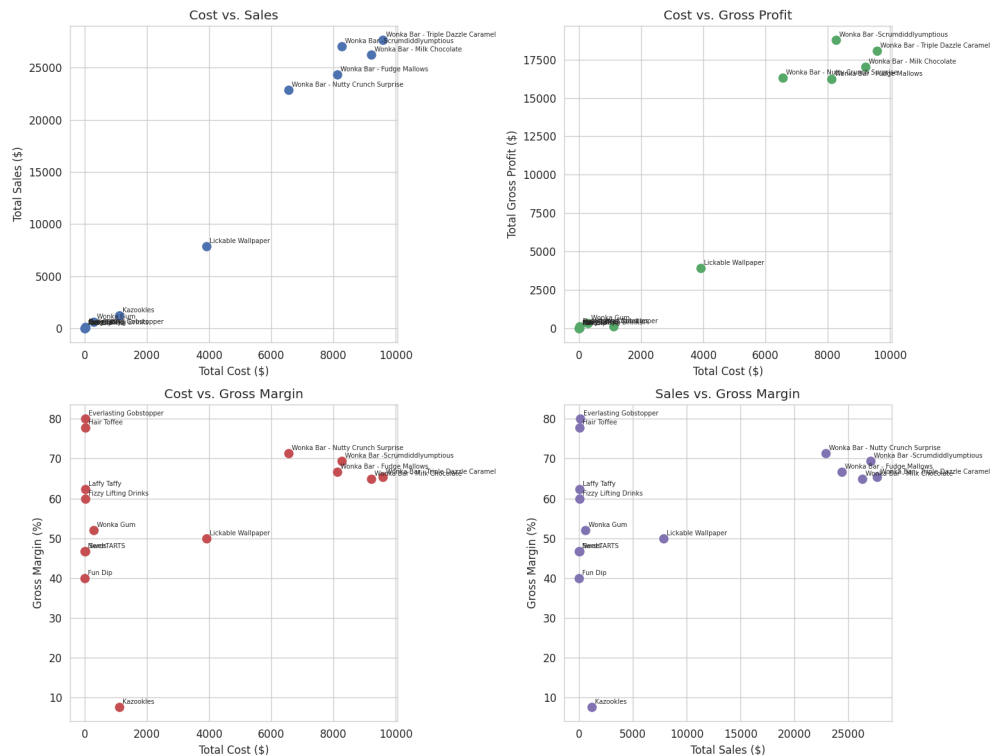


Figure 6. Cost vs. Sales, Cost vs. Gross Profit, Cost vs. Margin, and Sales vs. Margin, by product.

The five Wonka Bars cluster together with cost, sales, and profit scaling proportionally to one another — a sign of a consistent, healthy cost structure across the core product line. Kazookles is a clear outlier: moderate cost relative to an unusually low margin, distinguishing it from every other product in the diagnostic plots. These patterns are treated as review flags for a pricing or sourcing conversation, not as automatic verdicts.

4. Discussion — Key Findings

- Nassau Candy is a **chocolate-led business**: the Chocolate division accounts for 92.7% of revenue and 94.9% of gross profit from just five SKUs.
- Revenue and profit are both concentrated in 5 of 15 products (33.3% of the catalog) — the Wonka Bar line — indicating meaningful product dependency risk.
- **Kazookles** is the clearest margin-risk product in the catalog: real sales volume but a 7.7% gross margin, far below every other product.
- Chocolate converts revenue to profit slightly more efficiently than the company average; Other does so slightly less efficiently, driven by Kazookles.
- Several Sugar-division products carry high per-unit margins (up to 80%) but contribute negligible dollars — a minor part of the business as currently sold, and a directional opportunity rather than a proven driver.

5. Recommendations

- Investigate Kazookles' cost structure and/or pricing — its margin is a clear outlier relative to the rest of the catalog and is the most actionable single finding in this analysis.
- Given the concentration in five chocolate SKUs, prioritize supply, cost, and pricing stability for those products, since they carry the majority of company profit.
- Consider a volume or marketing push behind high-margin, low-volume Sugar products (e.g., Everlasting Gobstopper, Hair Toffee), since their profitability per dollar is already strong — this is a data-supported opportunity flag, not a guaranteed outcome, and would benefit from a small pilot before broader investment.

6. Limitations

- Ship Date data quality issues meant delivery-time or shipping-efficiency analysis could not be included; this was not part of the stated objective, but is worth revisiting if corrected data becomes available.
- The catalog is small (15 products), so segmentation thresholds are based on medians and quartiles across those 15 products rather than a larger statistical sample.
- Margin volatility is measured only across the two-year window present in the data; seasonality beyond that window cannot be assessed.
- No factory, coordinate, or shipping-route data exists in this dataset, so sourcing/logistics-side cost drivers could not be examined — only the sales-side financials.

7. Conclusion

This analysis shows that Nassau Candy Distributor's \$138,284 in recorded sales and \$91,076 in gross profit are driven overwhelmingly by a small set of chocolate products, with the rest of the catalog contributing comparatively little. Separating profitability from raw sales volume surfaced one clear, actionable finding — Kazookles' unusually weak margin — that would not have been visible from a sales-only view, along with a concentration risk worth monitoring as the product line evolves. The accompanying notebooks and interactive dashboard allow this analysis to be re-run and re-filtered as new transaction data becomes available.

Tools & Reproducibility

Analysis performed in Python (pandas, numpy) with matplotlib/seaborn for static charts and Plotly for the companion interactive Streamlit dashboard. Full methodology, code, and intermediate outputs are available in notebooks/01_EDA_and_Cleaning.ipynb, 02_Profitability_Analysis.ipynb, and 03_Advanced_Analysis.ipynb, with the interactive dashboard in app.py.